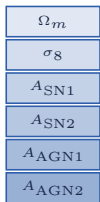


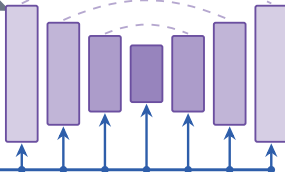
θ_{ask}
requested cosmology



noise x_T

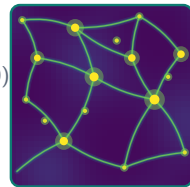


Conditional diffusion
UNet denoiser



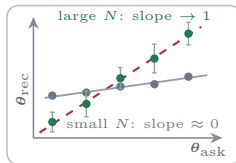
sample (DPM-50)

Generated HI map
 128×128

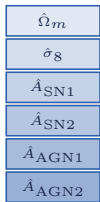


cross-attention: θ enters every level

calibration
 $\theta_{\text{rec}} \approx \theta_{\text{ask}} ?$



preprocess

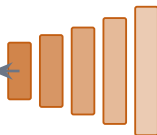


regress



MLP head $1024 \cdot 512 \cdot 256$

avg $\oplus$ max



* VGG16 encoder

$3 \text{ ch} \cdot 224^2$

θ_{rec}
recovered cosmology

diagnostic probe · trained on real slices only · frozen for generated fields